

Lesson 4 Sets

1. List the elements of the following sets:
 - a) $A = \{x \in \mathbb{Z} : |x| < 10 \text{ and } x \text{ is even}\}$
 - b) $B = \{x \in \mathbb{Z} : x^3 \leq 1\}$
 - c) $C = \{2n + 1 : n \in \mathbb{N}\}$
2. Write the following sets using set-builder notation.
 - a) $A = \{-1, 0, 1, 2, 3, 4, \dots\}$
 - b) $B = \{4, 9, 16, 25, \dots, 400\}$
 - c) $C = \{\dots, -8, -4, 0, 4, 8, 12, \dots\}$
3. Let $A = \{-1, 0, 1\}$.
 - a) Find $\mathcal{P}(A)$.
 - b) True or false:
 - i. $0 \in A$
 - ii. $0 \subseteq A$
 - iii. $0 \in \mathcal{P}(A)$
 - iv. $0 \subseteq \mathcal{P}(A)$
 - v. $\emptyset \in A$
 - vi. $\emptyset \subseteq A$
 - vii. $\emptyset \in \mathcal{P}(A)$
 - viii. $\emptyset \subseteq \mathcal{P}(A)$
4. Given the universal set $U = \{0, 1, 2, 3\}$ and the sets $A = \{0, 1\}$, $B = \{1, 2\}$, list the elements of the following sets.
 - a) $\overline{A}$
 - b) $A \cap B$
 - c) $A \cup B$
 - d) $\overline{A \cup B}$
 - e) $\mathcal{P}(A \cap B)$
 - f) $\mathcal{P}(A) \cap \mathcal{P}(B)$
5. Let $A = \{1\}$, $B = \{1, \{1\}\}$, $C = \mathcal{P}(A)$, and $D = \mathcal{P}(B)$.
 - a) List the elements of C and the elements of D .
 - b) True or false?
 - i. $A \in B$
 - ii. $A \subseteq B$
 - iii. $C \in D$
 - iv. $C \subseteq D$
6. Let A and B be any sets. Determine if each of the following statement is true.
 - a) $B \subseteq A$ if and only if $B \in \mathcal{P}(A)$
 - b) $A \in \mathcal{P}(A)$
 - c) $A \subseteq \mathcal{P}(A)$

7. Given the sets,

$$A = \{x \in \mathbb{Z} : x|2\}, \quad B = \{x \in \mathbb{N} : x|4\}, \quad C = \{x \in \mathbb{Z} : 2|x\}$$

List the elements of:

- a) $A \setminus B$
- b) $A \Delta B$
- c) $A \setminus (B \setminus C)$
- d) $(A \setminus C) \setminus B$
- e) $A \times B$
- f) $B \times A$

8. Given the sets,

$$A = \{0, 1, 2, 3\}, \quad B = [1, 2]$$

Draw the following sets on the real line.

- a) $A \cap B$
- b) $A \cup B$
- c) $A \setminus B$
- d) $B \setminus A$

Draw the following sets on the Cartesian plane.

- e) $A \times B$
- f) $B \times A$

9. Let $A = \mathbb{R} \setminus \mathbb{N}$. Determine the following sets.

- a) $A \cup \mathbb{Z}$
- b) $\mathbb{Q} \setminus A$

10. If A and B are subsets of $\mathbb{R}$, we may define sum and product of sets A and B :

$$A + B = \{a + b : a \in A, b \in B\}$$

$$A * B = \{ab : a \in A, b \in B\}$$

If $c \in \mathbb{R}$, we may also define

$$cA = \{ca : a \in A\}$$

- a) $\mathbb{N} + \mathbb{N} =$
 - A. $\mathbb{N}$
 - B. $\{2, 4, 6, 8, 10, 12, \dots\}$
 - C. None of the above
- b) $\mathbb{N} * \mathbb{N} =$
 - A. $\mathbb{N}$
 - B. $\{1, 4, 9, 16, 25, 36, \dots\}$
 - C. None of the above
- c) $2\mathbb{N} =$
 - A. $\mathbb{N} + \mathbb{N}$
 - B. $\{2, 4, 6, 8, 10, 12, \dots\}$
 - C. None of the above
- d) True or false: $-\mathbb{Q} = \mathbb{Q}$